CLOUD COMPUTING COURSE PROJECT

Quantifying V2I Channel Overhead from Continuous Digital Twin State Synchronisation

A SUMO and Python study of vehicular edge telemetry load

Roll number	Team member	Role
241CS135	Mudu Keerthana	Member
241CS144	R Trishna	Member
241CS152	Harika Seesa	Member
241CS236	Madhuri Nallaboyina	Team leader

Assignment: Task 1 - V2I channel overhead

Submission report draft for mentor review

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This report is generated from the project code and experiment CSVs. The channel assumptions and limitations are stated explicitly; the results require mentor approval before final submission.

Abstract

Vehicular digital twins rely on frequent vehicle-state updates sent over a shared vehicle-to-infrastructure (V2I) uplink. This project implements a reproducible SUMO mobility scenario and a first-order Python link-budget model to estimate telemetry airtime, packet collisions, packet delivery ratio (PDR), and the remaining channel resource for task offloading. Five vehicle densities (5, 10, 20, 50, and 100), three update intervals (50, 100, and 500 ms), and five random-phase repetitions were evaluated. With a 32-byte payload and the assigned 5.9 GHz, 10 MHz radio parameters, the highest baseline utilisation was 2.15% at 100 vehicles and a 50 ms update interval. The required 60% saturation threshold was not reached in the tested range. A 64-byte, path-loss-exponent-3.5 sensitivity case reached 9.21% at the same density and interval.

1. Problem and objectives

The assigned problem asks how much shared V2I uplink capacity is consumed by continuous digital-twin state synchronisation as vehicle density increases. The project objectives are to generate SUMO vehicle traces; estimate received power, SNR, and Shannon capacity; simulate periodic telemetry and overlapping transmissions; report utilisation, PDR, residual bandwidth, per-vehicle update rate, and the first tested density above 60% utilisation.

2. Tools and implementation

SUMO 1.28.0 supplies a 1,000 m two-lane corridor and TraCI controls vehicle insertion and state collection. Python 3.14.2 is used with NumPy for numerical summaries and Matplotlib for plots. The code is separated into mobility trace generation (src/trace_mobility.py), link-budget functions (src/channel_model.py), experiment execution (src/run_experiments.py), and plotting (src/plot_results.py). Unit checks are in tests/test_channel_model.py.

3. Mobility scenario

Setting	Value
Road	Straight 1,000 m corridor; two lanes; 20 m/s speed limit
Simulation duration and step	10 s per scenario; 0.05 s step
Vehicle densities	5, 10, 20, 50, 100
Vehicle start positions	Evenly spaced from 100 m to 800 m
Initial speed and random seed	5 m/s; seed 42
Mobility output	Position, speed, heading, acceleration; 37,000 rows

4. Radio and traffic model

Parameter	Baseline value
Carrier frequency	5.9 GHz
Channel bandwidth	10 MHz
Transmit power	20 dBm
Thermal noise PSD / receiver noise figure	-174 dBm/Hz / 9 dB
Integrated receiver noise over 10 MHz	-95 dBm
Reference distance and path-loss exponent	1 m / 3.25
Free-space reference loss at 5.9 GHz	About 47.87 dB
Logical RSU positions	250 m, 500 m, 750 m
Payload / upload intervals	32 bytes / 50, 100, 500 ms

The model uses three logical RSU positions along the road. Every vehicle associates with the nearest RSU, but all vehicles contend on one shared logical 10 MHz channel; no spatial reuse between RSUs is assumed. Distance is the longitudinal road-coordinate difference.

Link budget

For distance d at or beyond the 1 m reference point, path loss is PL(d) = PL(1 m) + 10 alpha log10(d / 1 m). Reference loss is the free-space loss 20 log10(4 pi f d0 / c). Received power is transmit power minus path loss. Receiver noise is N = -174 + 10 log10(B) + NF dBm, giving -95 dBm at 10 MHz with a 9 dB noise figure. Linear SNR is 10^(SNR_dB/10), and link capacity is C = B log2(1 + SNR_linear). Ideal payload airtime is payload bits divided by C.

Traffic, collisions, and metrics

For each run, vehicle upload phases are independently drawn from a uniform distribution over one upload interval and then remain fixed for that run. Five seeded phase repetitions are pooled. A collision occurs when two packet airtime intervals overlap anywhere on the shared channel; both packets are counted as unsuccessful. Positions between the 50 ms SUMO samples are linearly interpolated. The effective Shannon capacity is total payload bits divided by total ideal airtime.

Metric	Definition
Offered load	Generated payload bits / simulation duration
Utilisation	Sum of ideal packet airtimes / simulation duration
Busy time	Union of packet airtime intervals / duration
PDR	(Generated packets - collided packets) / generated packets
Residual bandwidth	10 MHz x max(0, 1 - utilisation fraction)
Saturation threshold	First tested density with utilisation > 60%

Offered utilisation counts overlapping attempts separately and may exceed 100% under overload; busy time counts overlapped wall-clock time once. Residual bandwidth and capacity are clipped at zero.

5. Baseline results

The following measurements are for the maximum tested density of 100 vehicles. Packet counts and PDR pool five random-phase repetitions. The nominal per-vehicle transmission rates are 20, 10, and 2 telemetry updates per second for the 50, 100, and 500 ms intervals.

Update interval	PDR	Utilisation	Residual bandwidth
50 ms (20 updates/s/vehicle)	96.09%	2.15%	9.79 MHz
100 ms (10 updates/s/vehicle)	97.96%	1.07%	9.89 MHz
500 ms (2 updates/s/vehicle)	99.74%	0.21%	9.98 MHz

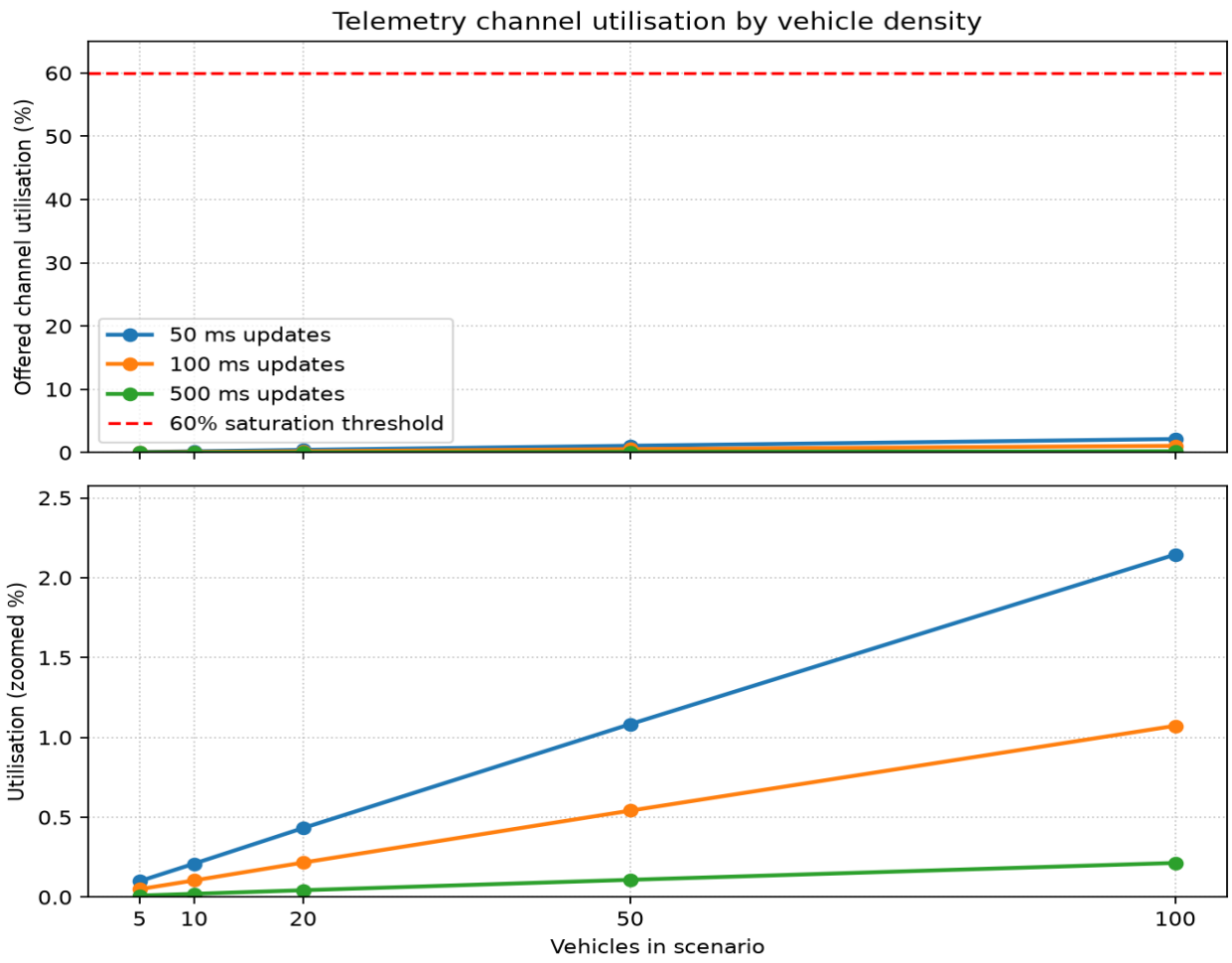
Saturation finding

The 60% utilisation threshold was not reached for any tested update interval through 100 vehicles. This is a bounded result for the stated payload, channel model, and density set; it does not establish a threshold beyond 100 vehicles.

Update interval	60% threshold result
50 ms	not reached up to 100 vehicles
100 ms	not reached up to 100 vehicles
500 ms	not reached up to 100 vehicles

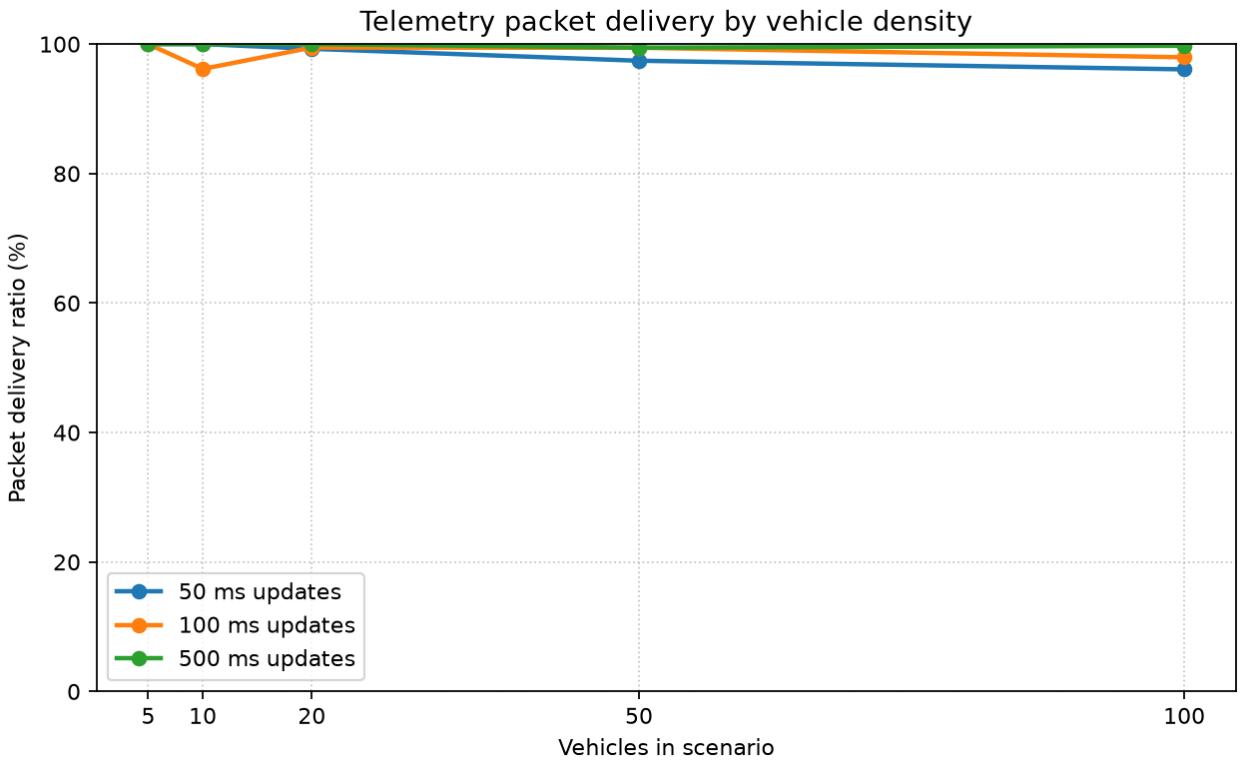
At 100 vehicles, the baseline effective Shannon capacity is about 23.86 Mbps for the 50 ms case, while the offered telemetry load is 0.512 Mbps. The resulting low airtime load explains why the assigned 60% threshold is not crossed under the idealised model.

6. Utilisation by density and update interval



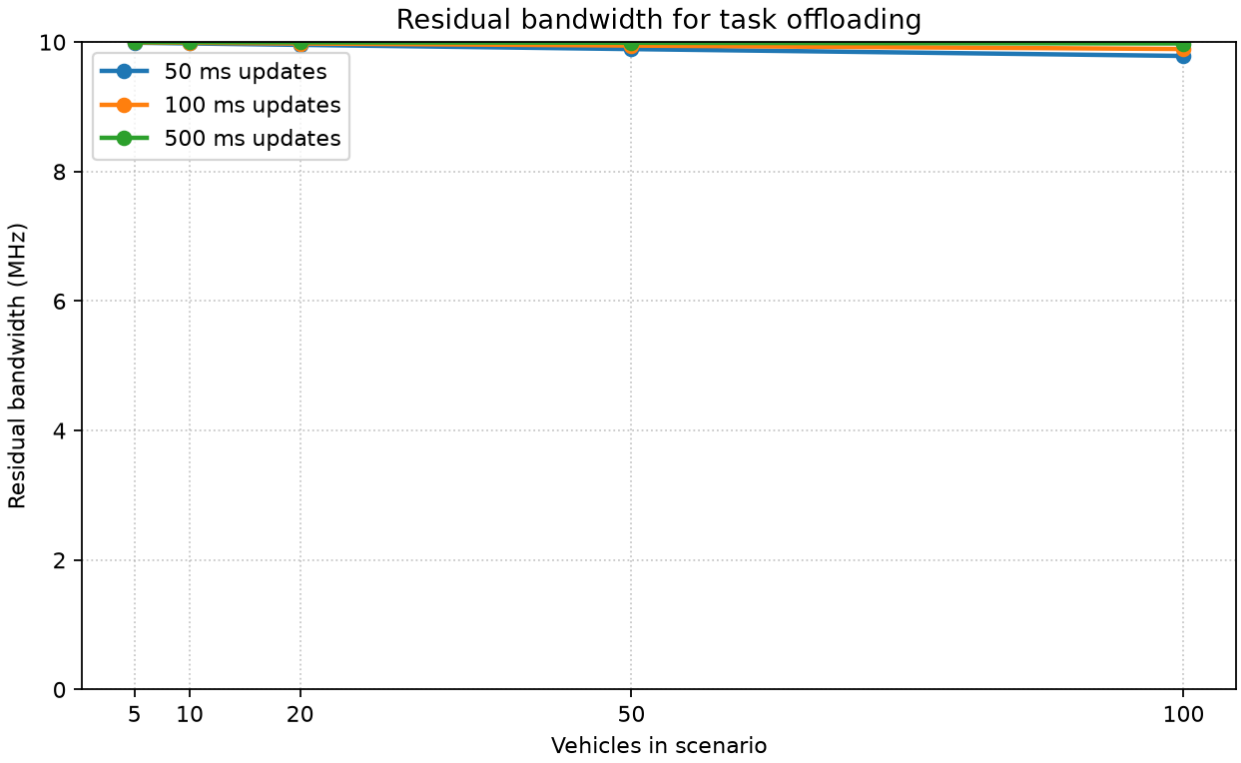
The upper panel places the assigned 60% threshold in context; the lower panel zooms in on the measured baseline range.

7. Packet delivery ratio



PDR falls as density and update frequency increase because overlapping transmissions collide in the shared-channel model.

8. Residual bandwidth for offloading



Residual bandwidth is the 10 MHz logical channel after offered telemetry airtime has been subtracted.

9. Sensitivity analysis

To test a higher telemetry burden, payload size was increased to 64 bytes and the path-loss exponent to 3.5, keeping the same mobility traces, density set, channel bandwidth, and five repetitions. At 100 vehicles:

Update interval	Utilisation	PDR	Residual bandwidth
50 ms	9.21%	82.76%	9.08 MHz
100 ms	4.61%	91.99%	9.54 MHz
500 ms	0.92%	98.40%	9.91 MHz

Even in this higher-overhead tested case, utilisation remains below 10% at 50 ms and below the required 60% threshold. Collision losses can still reduce PDR even when average channel utilisation is modest; the uncoordinated periodic access and shared-channel assumptions make PDR sensitive to upload phase alignment.

10. Discussion and limitations

The observed trends match the model: more vehicles and shorter update intervals increase offered telemetry airtime, reduce residual bandwidth, and create more opportunities for packet overlap. At the same 100-vehicle density, a 50 ms interval generates ten times as many updates per vehicle as a 500 ms interval. In the baseline, PDR is 96.09% at 50 ms and 99.74% at 500 ms.

This model is not an 802.11p/DSRC simulator. It uses Shannon-limit rates and payload-only airtime; it omits MAC/PHY headers, CSMA/CA backoff, coding-rate limits, fading, shadowing, receiver sensitivity, cross-RSU interference, retransmissions, and task-offloading packets. There is no independent PHY outage/drop threshold because none was specified in the assignment; the PDR loss counted here is from overlapping transmissions. Accordingly, the results are an optimistic, reproducible baseline and should not be presented as measured DSRC performance.

11. Conclusion

A working SUMO/TraCI-to-Python pipeline now generates repeatable mobility traces and evaluates telemetry overhead across all assigned densities and update intervals. Under the stated 32-byte baseline, the maximum measured offered utilisation is 2.15%, so no tested density reaches 60%. The 64-byte / exponent-3.5 sensitivity case increases maximum utilisation to 9.21%, still below the threshold. The next improvement should be a mentor-approved PHY/MAC model including packet overhead and realistic access control, followed by re-running and comparing these baseline results.

Reproduction

Build the road with netconvert, then run `python src\trace_mobility.py`, `python -m src.run_experiments`, and `python -m src.plot_results` from the project directory. The default experiment uses seed 42 and five phase repetitions; generated CSVs and figures are retained under data/ and results/.

References

[1] Course-assigned Task 1 problem statement, V2I digital-twin synchronisation overhead, 2026.

[2] C. E. Shannon, 'A Mathematical Theory of Communication,' Bell System Technical Journal, vol. 27, pp. 379-423 and 623-656, 1948.

[3] H. T. Friis, 'A Note on a Simple Transmission Formula,' Proceedings of the IRE, vol. 34, no. 5, pp. 254-256, 1946.

[4] German Aerospace Center (DLR), Eclipse SUMO Documentation, https://sumo.dlr.de/docs/ (accessed October 2026).

Mentor approval: _____

This is a draft prepared for review. Update it if the mentor changes the radio assumptions or approves a different model.